

LENZ'S LAW AND INDUCED EMF

L7

Statement

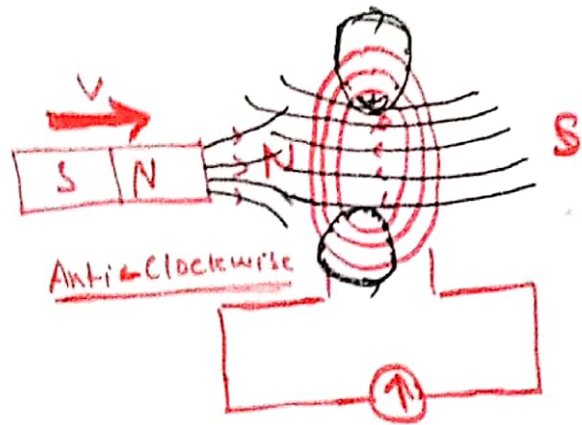
"Direction of induced current is always so as to oppose the change which causes the current"

Explanation

- * In 1834 Heinrich Lenz discovered this law
- * It refers to induced current Not on induced emf, so it is only applied on close loops or coils
- * Direction of induced emf can be found by considering the direction of induced current
- * Let's apply Lenz's law to the coil in which current is induced by the movement of bar magnet
- * we know current carrying coil will act as a Bar Magnet one face will act as North pole "N" and other will act as South pole "S"

② When Bar Magnet is Moved Towards coil

(i) when the Bar Magnet is Moved Towards the coil It will Increase the flux through the coil



(ii) according to lenz's law Direction of Induced current will be such that it will oppose the Increasing flux, so, the induced current will try to Decrease the flux

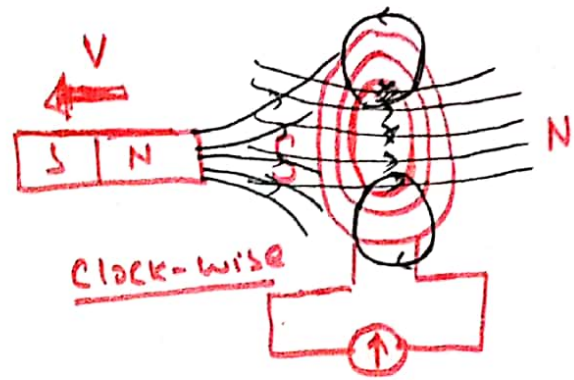
(iii) To Do so Direction of Induced current will be Anti-clock wise, and face of the coil Towards the magnet will become a North pole ^{'N'} and oposit face will become South pole ^{'S'} (Right hand Rule)

(iv) In this way Induced current will oppose the Change (Increasing flux), which is producing it.

“ push of the magnet is the change that produces the Induced current and Induced current acts to opose the push”
~~LENZ'S LAW and Induced EMF~~

⊗ Bar Magnet is Moved Away from coil

i) When the Bar Magnet is moved away from the coil it will decrease the flux through the coil



ii) According to Lenz's Law Direction of Induced current will be such that it will oppose the Decreasing flux, so the induced current will try to increase the flux

iii) To do so, Direction of induced current must be clock-wise, and face of the ~~magne~~ coil towards the magnet will become a South pole ^{'S'} and opposite face will become North pole (N)

iv) In this way induced current will oppose the change (~~Decreasing~~ ^{Decreasing} flux) which is producing it

⊗ pull of the Magnet will generate an induced current and this current will oppose the pull of the Magnet.

LENZ'S LAW AND CONSERVATION OF ENERGY

i) Consider an experimental setup as shown in figure

ii) When rod will move ^{forwards Right} in magnetic field an emf will be induced in it

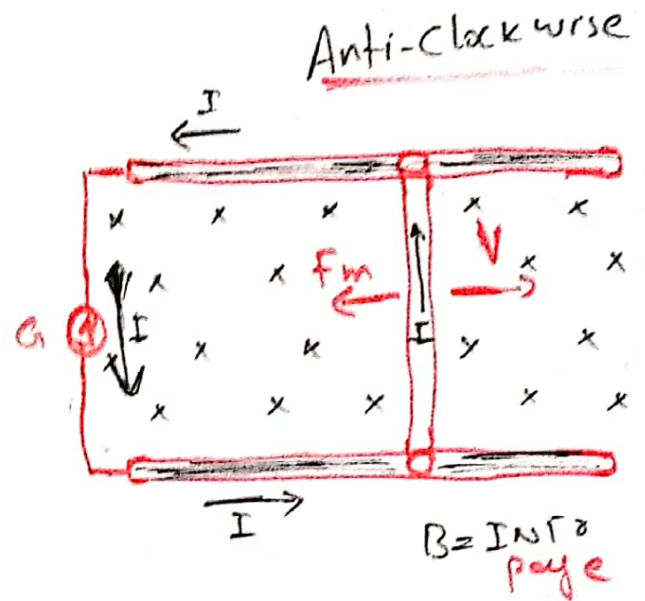
iii) A current will start flowing through the loop in Anti-clockwise Direction

iv) As, we know, a force will act on the conductor moving in magnetic field given as

$$F_m = I L B \sin 90^\circ$$

v) By Right hand Rule we can find easily that this force is acting opposite to that of Velocity 'v' of Rod, so it tends to stop the rod

vi) To keep Rod moving an external Dragging force equal to F_m in magnitude must be in opposite Direction is required to keep rod moving with constant Velocity

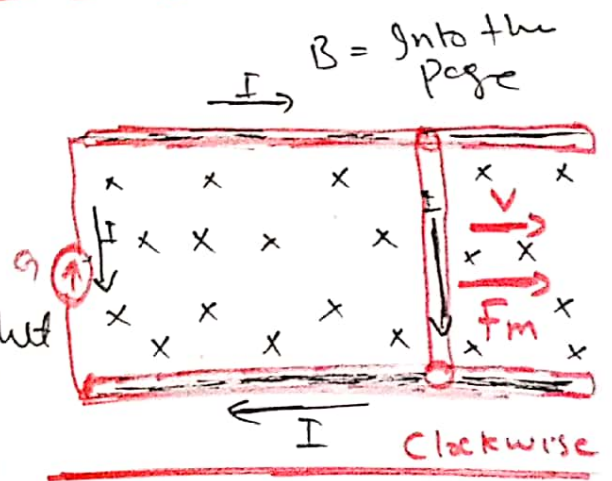


vii) This dragging force is providing Energy for the Induced current to flow

viii) Thus we can say Electromagnetic Induction is exactly according to Law of Conservation of Energy

If INDUCED current IS clock-wise

③ If the Induced Current is in Clockwise Direction, then from figure, By using Right hand Rule we can easily found that ' F_m ' would be in the Direction of Velocity



* It will induce a stronger current, and hence stronger Magnetic field, which will increase Magnetic Force (F_m) further

Motion of the rod will accelerate more and more

* So, we are obtaining increasing K.E of motion from Nowhere, so consequently process becomes Self accelerating, which is Against the Law of Conservation of Energy